# COLD PAK 100

# INTRODUCTION MANUAL TO THE CP-100

**Manual version 1.0 (31-March-2004)**


**C/ ZUAZNABAR 36**

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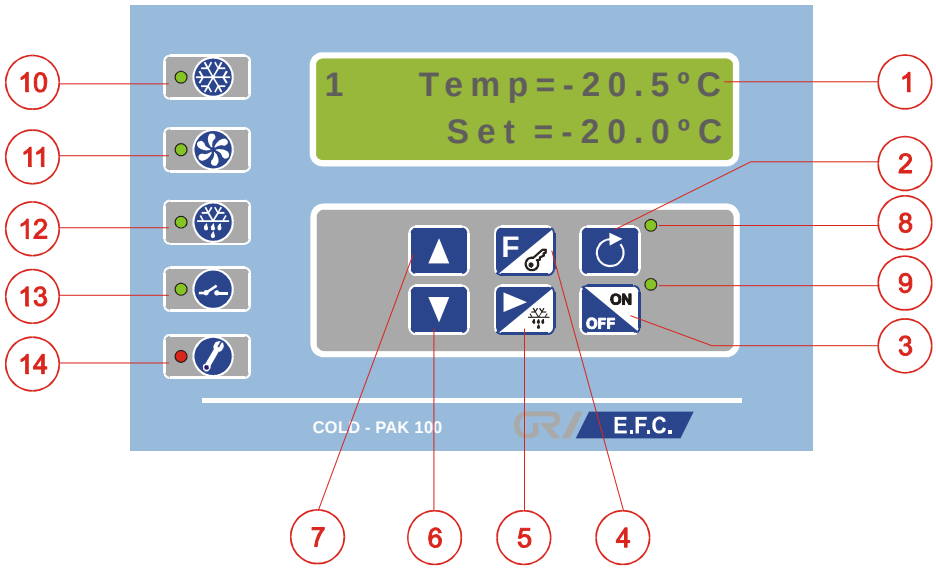
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# 1. DESCRIPTION OF THE COLD-PAK 100 REMOTE CONTROL FRONT PART




1) THE LCD DISPLAY

The LCD Display consists of 2 lines and each one has 16 characters of good visibility. This display indicates the real Temperature and Relative Humidity figures (if there is humidification control) and their respective set-points. It also shows the situation of some events, such as errors, and gives different messages during the configuration.

2) KEY 'FOR ALL'

This key is located in the "Configuration Menu". It allows to save the modified figures, just in one Control or in all Controls at the same time. The same function can be carried out with the Key 'ON/OFF'.

If we press this key for one second, it can also *start / stop* the scanner of the Controls, which are installed in its communication net. If we press this key shortly, it shows the output relays figures, situation of the machine, temperature, and other data of another Control.

The time of the Control's display is by default of 10 seconds. This time can be modified by the user. When the Key 'FOR ALL' is activated (Led num. 8 illuminated), the COLD-PAK 100 REMOTE CONTROL will select the Control, which is indicated on the display. Once the time, which has been configured by the user, is over, another Control will be displayed.

3) KEY 'ON/OFF'

This Key allows to start / stop the Controls. If the Key 'FOR ALL' is not activated, the only control that should start / stop, would be the one which is displayed by the remote control. If the Key 'FOR ALL' is activated and press the Key , all the Controls can start / stop.

The Controls will not start / stop, if there is the following indication: 'Modifications = Not Allowed.'

4) KEY 'FUNCTION'

This key allows to display the temperature figures or the humidity ones. If we press it for 2 seconds, it ask for the access password to go into the configurations menu of the COLD-PAK 100

REMOTE CONTROL or the Controls. If we press it for 7 seconds, it allows to go into the configurations / maintenances menu, but modifications are not allowed.

5) KEY 'INPUT' 

This key is used to edit and validate (saved in memory) the temperature set-point, the humidity set-point and all configuration parameters. If we press it for 2 seconds, it allows to start / stop the defrosting of the Control, that is displayed at that moment.

The defrosting can not start / stop, if there is the following indication: 'Modifications = Not Allowed'.

6) KEY 'GO DOWN' 

This key allows us to move down along the different configuration menus and also to decrease the edited figures.

7) KEY 'GO UP' 

This key allows us to move up along the different configuration menus and also to increase the edited figures.

8) LED 'FOR ALL'

- If the Led is illuminated= the Key 'For All' is activated. It also means, that the remote control is scanning the Controls.
- If the Led is not illuminated= the Key 'For All' is not activated
- If the Led blinks= the key 'For All' is activated, but all the Controls have not the same set-points or they have not the same ON/OFF situations.

9) LED 'START /STOP'

It indicates that the Control, which is being displayed, is in 'start/stop' situation.

10) LED 'COOLING ACTION IS WORKING'

The cooling action takes place when the VSL relay or AIX1 relay (depending on the configuration parameters) are activated.

11) LED 'FAN IS WORKING'.

This led indicates that the evaporator fan is working.

12) LED 'DEFROSTING ACTION IS WORKING'

This led has got an intermittent light, when the defrosting of the displayed Control is ready to start the defrosting action, but there is already one Control, which has started its defrosting.

13) LED 'AUXILIARY ACTION IS WORKING'

This led is on, when there is an auxiliary action. It takes place, when there is a cooling control output or humidity control output (depending on the settled configuration).

14) LED 'FAILURE'

The led is illuminated, when there is a serious failure, that activates the alarm (the 'OK' relay is not activated).

The led's light will be intermittent, when there is a failure, but this one has not activated the alarm ( the 'OK' relay is not activated).

If the led has an intermittent light, but the LCD does not show any failure /warning message, that means that the failure /warning has taken place in another Control. In this case, we will press the key  , until we will be able to see the message of failure/warning of the damaged Control.

# 2. USER'S EDITION

## 2.1. DESCRIPTION OF THE USER'S EDITION

When we are in the initial situation of the display, if we press shortly the key , we will go into the set-point's edition menu. The set-point's signal will be intermittent.

If there is this symbol (*) beside the Control number, that means, that the value of the set-point that is being edited, is not the same in all the Controls.

The keys or allow to modify the set-point's value. We press the key , in order to save the new value.

Modifications will not be possible, when there is the following indication: 'Modifications: Not Allowed".

**Edition of the temperature set-point:**

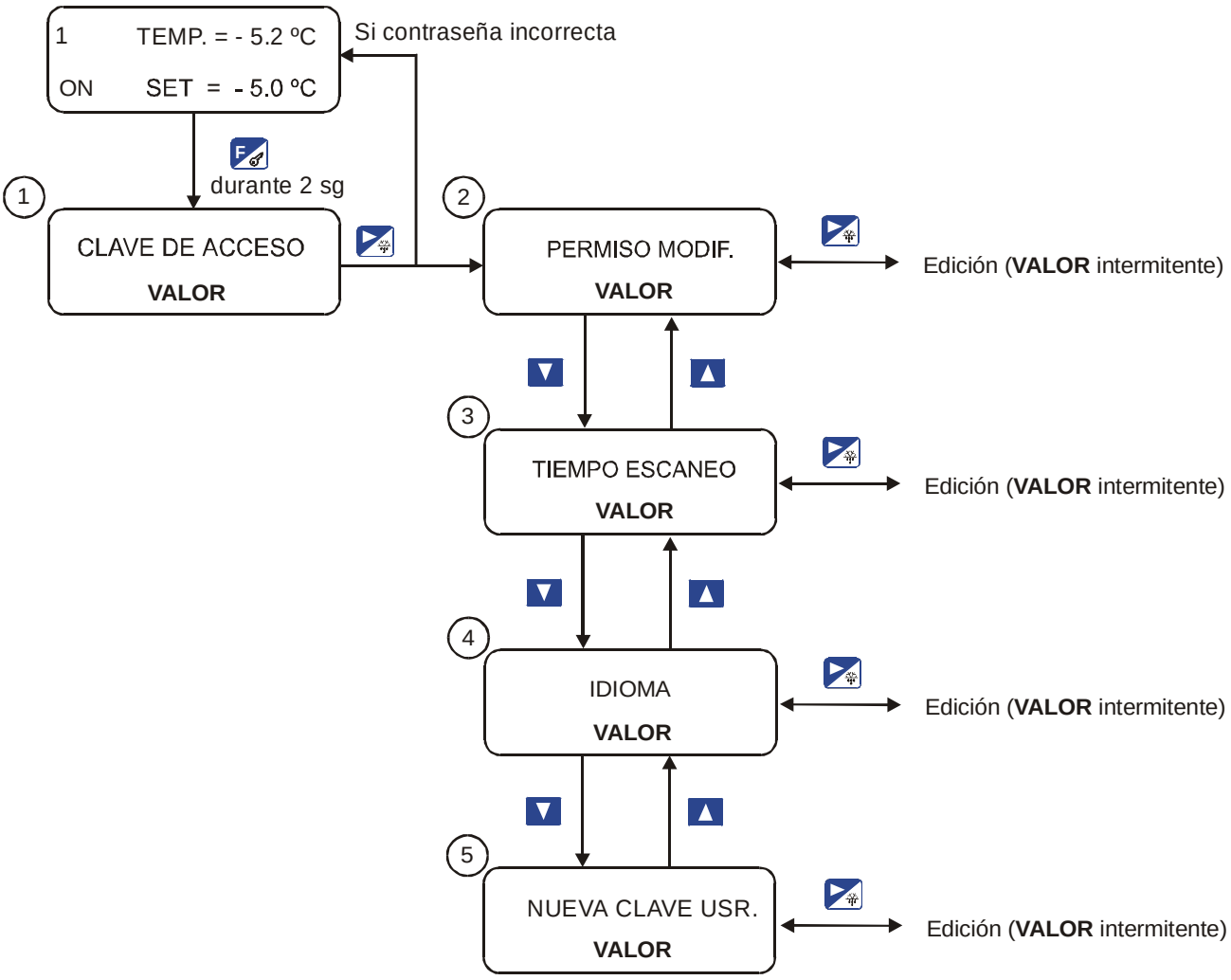
Only EFC is allowed to settle and modify the maximum and minimum set-points. The user can configure the values inside the max and min. set-points. ('LIM.MIN.CONSIGNA' and 'LIM.MAX.CONSIGNA') (see page 11).

**Edition of the humity set-point:**

Its values go from 0.0% to 100.0%.

# 3. USER'S CONFIGURATION OF THE COLD-PAK 100 REMOTE CONTROL

## 3.1. STRUCTURE OF USER'S CONFIGURATION




Pulsando la tecla se entra en la edición del **VALOR,** mostrándose éste de forma intermitente.
Con las teclas de y se modifica el valor
Para guardarlo habrá que pulsar la tecla , momento en el que dejará de parpadear.

Para salir de la configuración en cualquier momento y no guardar el valor modificado pulsar la tecla , o

If we press the key¿?, we go into the VALUE'S edition. The value's signal will be intermittent. The keys ¿? And ¿? Allows us to modify the value.

Press key to save the modified value. At that moment, the value's signal blinks no more.

Press keys... to leave the configuration menu at any time, without saving the modified value.

## 3.2. DESCRIPTION OF THE USER'S CONFIGURATION

In the main screen, if we press for 2 seconds the key F, we enter in the first configuration screen:

1) ACCESS CODE: Access code to the configurations menu

The keys ▲ and ▼ allow to introduce the value of the access code.

Once the value has been introduced, we should press the key ❄ to confirm it. If the value of the introduced code is not correct, the unit will leave the configurations and return to the main screen.

2) MODIF.PERMIT

This parameter can have 2 values: 'Yes' or 'No'. It is used to avoid that other users can make possible values' modifications.

If we select the value 'No':

. The set-points can not be modified,

. The manual start / stop defrosting can not be made

. The situation of the key ON/OFF can not be changed

. The order for outside cold inlet will be ignored.

3) TIME FOR SCANNING

In the main screen, when the key ⟳ is activated, this parameter indicates the time that the Cold-Pak 100 takes in order to go through each Control's display. The values are situated between 0 and 15 seconds. Please, bear in mind that if only one Control is connected to the remote control, there will not be any scan and this parameter will be ignored by the Cold-Pak 100.

4) LANGUAGE

This parameter indicates the language of the display.

5) NEW USER'S CODE. : New user's access code

The value of the new access code is introduced.

| | |
|---|---|
| Modifications permit | Yes |
| Time for scanning | 10 seconds |
| Language | English |
| User's configuration code | 235 |

# 4. EFC CONFIGURATION OF COLD-PAK 100 REMOTE CONTROL

## 4.1. REMARKS ABOUT THE USE OF THE KEYS DURING THE EDITIONS

The user can move himself through the configuration menu with the following keys:

- ▲ o ▼ in order to move inside the same level or to modify a value
- in order to go to a deeper level or to edit a value and confirm it
- in order to go up the following higher level or to escape without saving the value, that has being edited.
- to go to another Control or to activate or not activate the option 'For all', if we press the key for 1 second.
- in order to leave definitely the configuration

Once we are situated in one specific parameter of the screen, if we press the key we go into the edition of a value. The value's signal should blink. The value can be modified if we press the keys ▲ or ▼. In order to validate it and save it, press again the key .

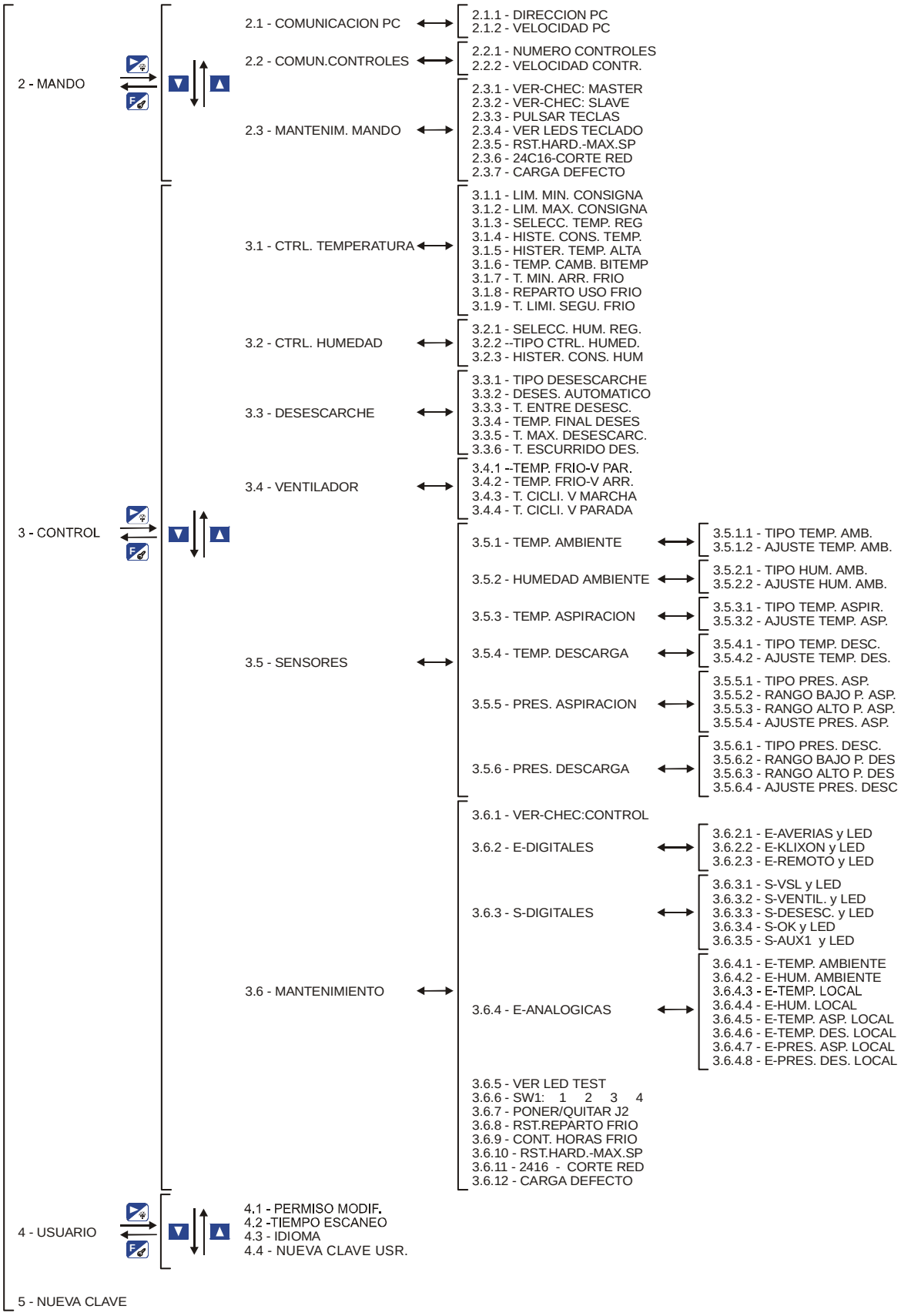
If the display shows the symbol (*) beside the number of the control, that means that the value is being edited, is not the same in all the Controls.

## 4.2. ACCESS TO CONFIGURATION MENU

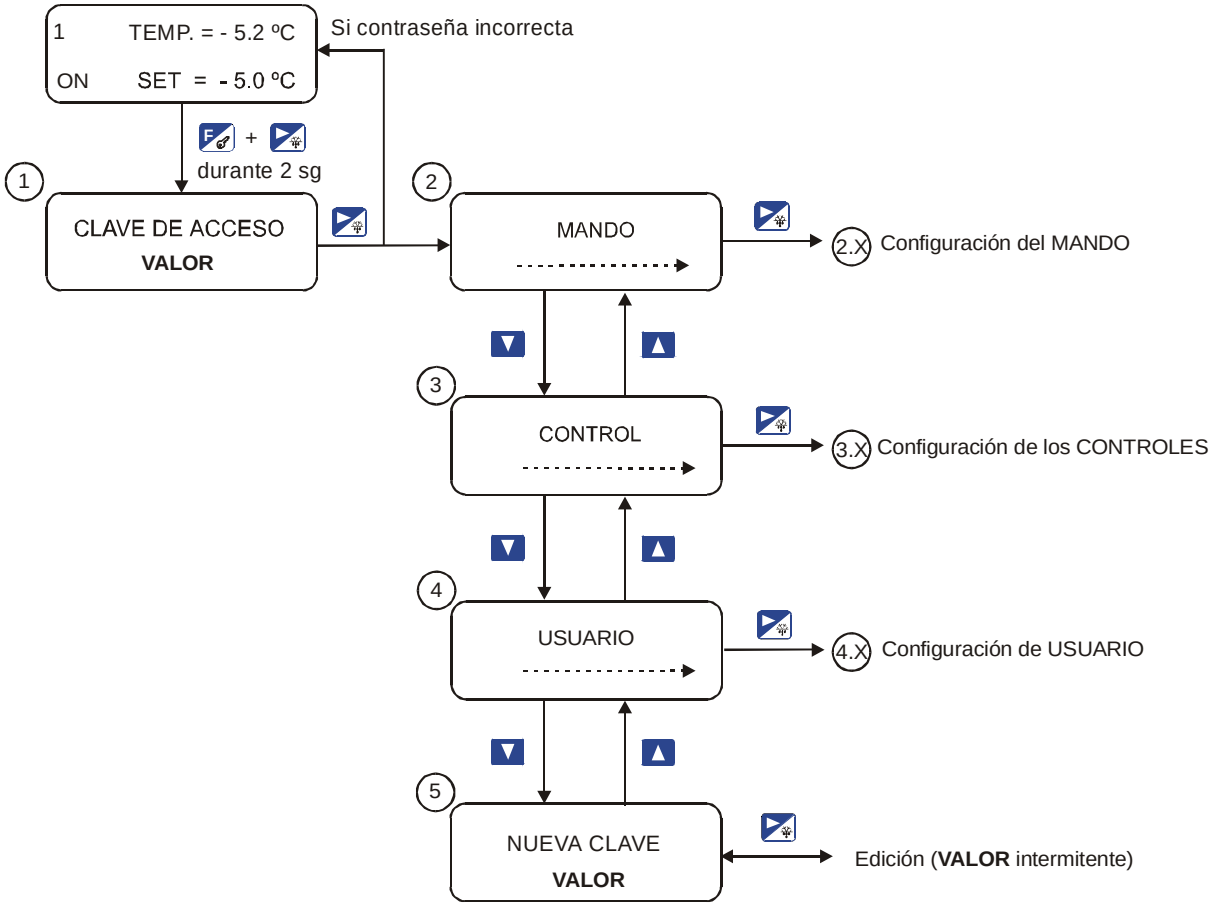
Please follow the following instructions in order to access to the configuration menu: in the main screen, press keys + for 4 seconds. In the access code screen, in the left lower part of the display will be shown a value in brackets. This value can have up to three figures: X Y Z. The user should communicate those figures to EFC. Then EFC will calculate the access code value, which is valid for this precise moment.

There is also the possibility of making a direct configuration without access code. In this situation, the user can see whatever he wants, except the access codes. There is not possible to make any modification. To go into in this possibility, press the key 'F' for 7 seconds.

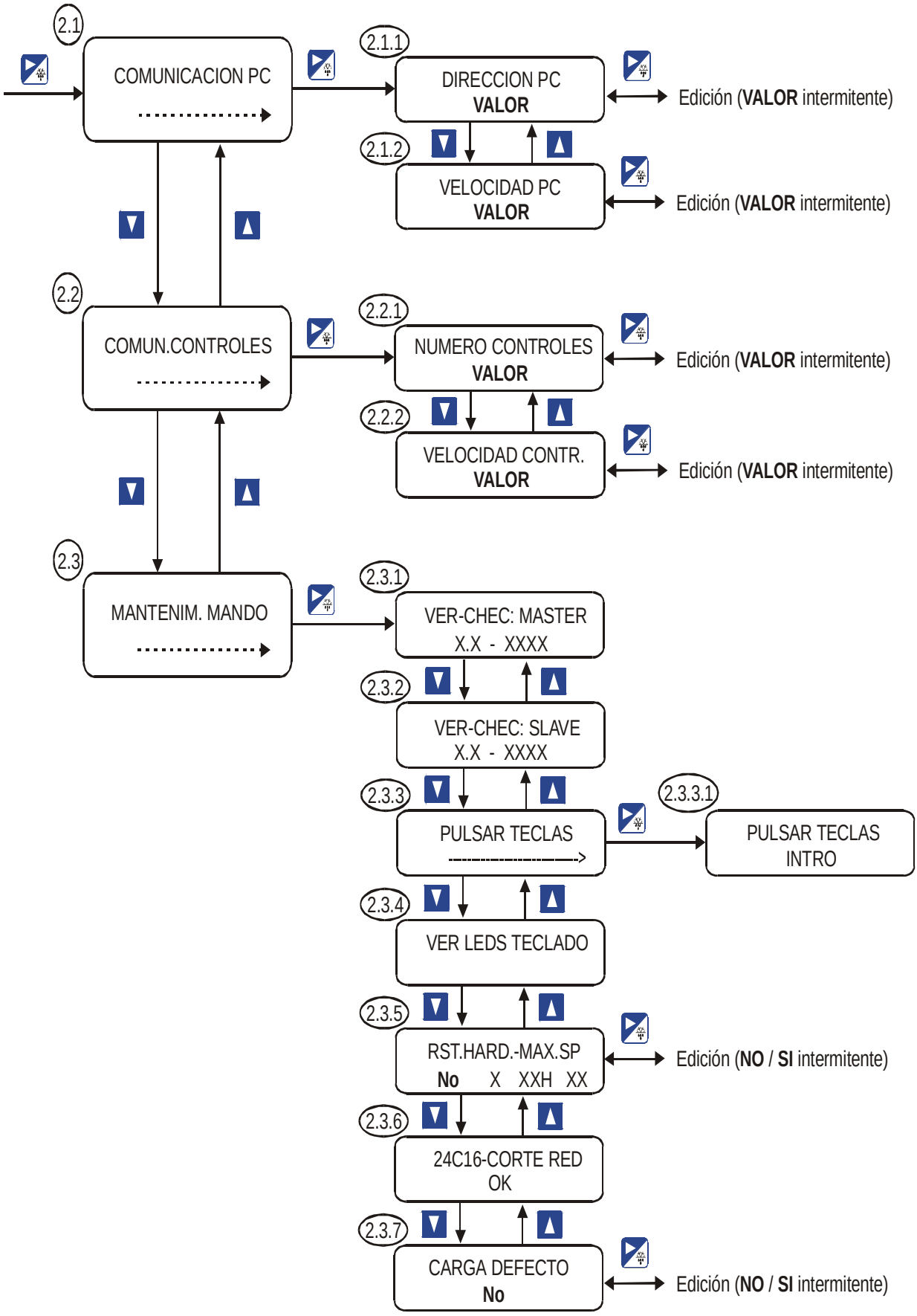
## 4.3. STURCTURE OF EFC CONFIGURATION



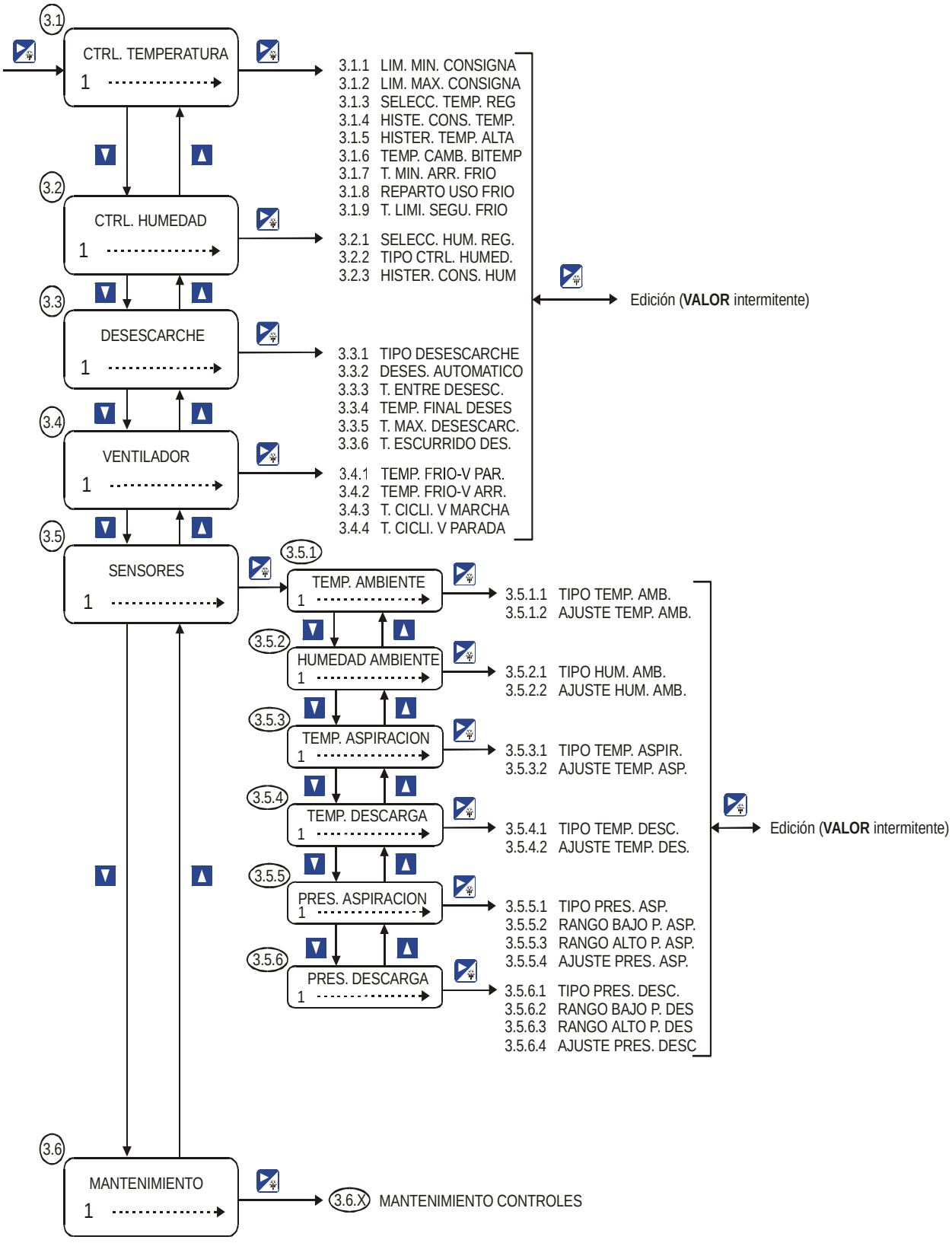
### 4.3.1. ***ACCESS TO THE CONFIGURATION:***



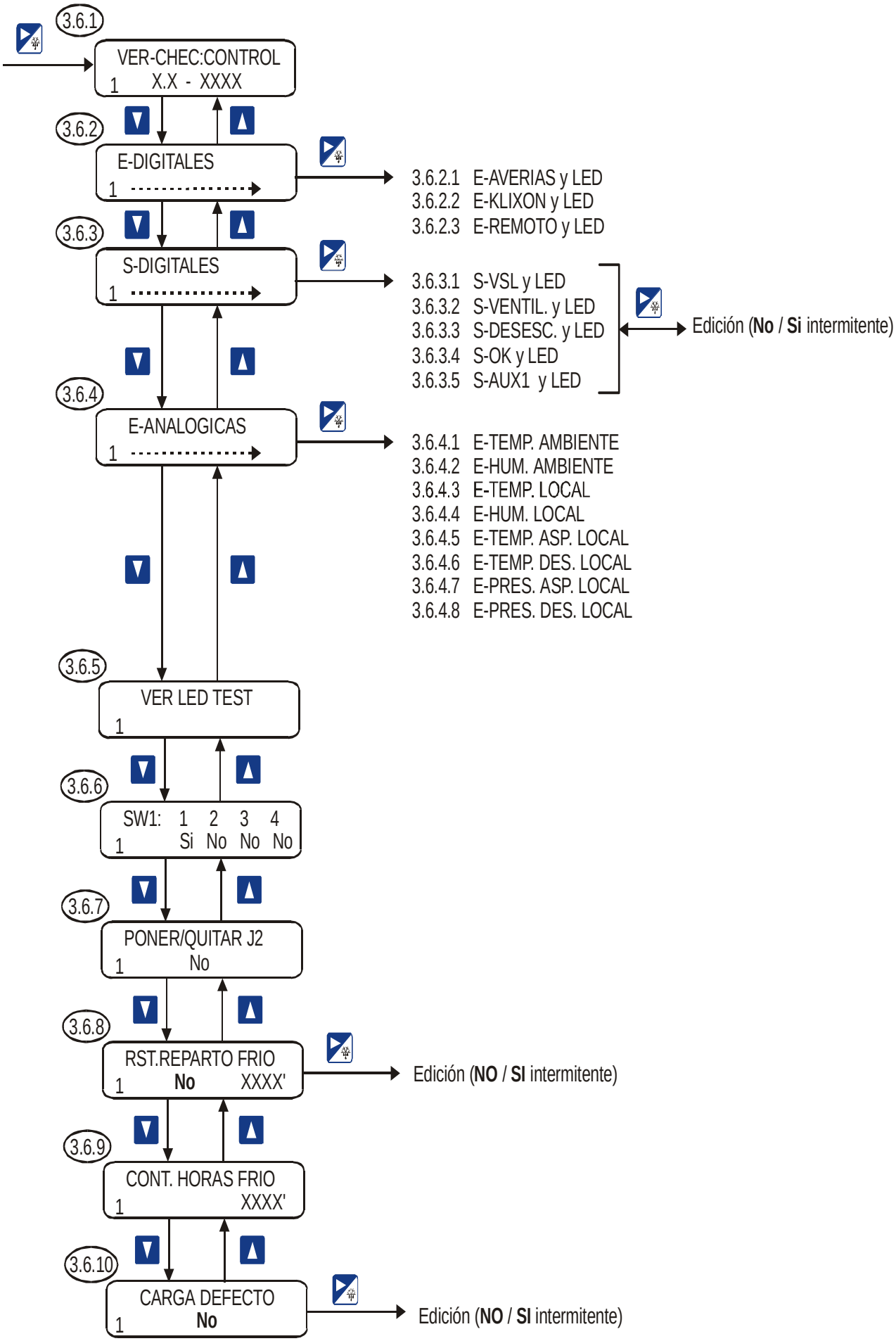
### 4.3.2. REMOTE CONTROL'S CONFIGURATION :



### 4.3.3. *CONTROLS' CONFIGURATION :*



### 4.3.4. CONTROLS' MAINTENANCE :



## 4.4. DESCRIPTION OF EFC CONFIGURATION

In the main screen, if we press for 2 seconds the keys  and  , we go into the first configuration screen:

1) **REMOTE CONTROL**:

Remote Control's values by default:

| | | | |
|---|---|---|---|
| 2.1.1 | DIRECCION PC | PC's address | 255 [1...255] |
| 2.1.2 | PC SPEED | Communication's speed with PC | 9600 bps |
| 2.2.1 | NUMBER OF CONTROLS | Number of controls | 4 [1...4] |
| 2.2.2 | CONTROL SPEED | Communication's speed with Controls | 9600 bps |

2) **CONTROL**:

Control's values by default:

| | |
|---|---|
| Contol's State | ON |
| Temperature Set-point | -10.0 °C |
| Humidity Set-point | 75.0 % |

| | | | |
|---|---|---|---|
| 3.1.1 | LIM.MIN.CONSIGNA | Minimum limit of temperature set-point's edition | -25 ºC [-45...20] |
| 3.1.2 | LIM.MAX.CONSIGNA | Maximum limit of temperature set-point's edition | 0 ºC [-45...20] |
| 3.1.3 | SELECC.TEMP.REG | Temperature selection | control nº1 → Main ...[Remote...Local]<br>rest of controls → Remote |
| 3.1.4 | HISTE.CONS.TEMP | Temperature set-point's hysteresis | 2.0 ºC [0,1...10] |
| 3.1.5 | HISTER.TEMP.ALTA | High temperature hysteresis | 5 ºC [2...20] |
| 3.1.6 | TEMP.CAMB.BITEMP | Temperature for changing into multi-temperature | -50 ºC [-50...-1] |
| 3.1.7 | T.MIN.ARR.FRIO | Minimum time between cool starts | 20 seconds [2...120] |
| 3.1.8 | REPARTO USO FRIO | Distribution of the compressors' wear | Sí |
| 3.1.9 | T.LIMI.SEGU.FRIO | Security limit time between cool starts | 15 minutes [5...120] |
| 3.2.1 | SELECC.HUM.REG. | Humidity selection | Local...[Main...Remote] |
| 3.2.2 | TIPO CTRL..HUM | Humidity control type | Humidified...[Do not exist...Dried] |
| 3.2.3 | HISTER.CONS.HUM | Humidity set-point's hysteresis | 4.0 % [1...10] |
| 3.3.1 | TIPO DESESCARCHE | Defrosting type | Gas...[Do not exist...Air...Electric] |
| 3.3.2 | DESES.AUTOMATICO | Automatic defrosting | Yes |
| 3.3.3 | T.ENTRE DESESCARC | Time between defrostings | 6 hours [1...24] |
| 3.3.4 | TEMP.FINAL DESES | Final defrosting temperature | 20 ºC [0...40] |
| 3.3.5 | T.MAX.DESESCARC | Maximum defrosting time | 30 minutes [1...60] |
| 3.3.6 | T.ESCURRIDO DES. | Drained defrosting time | 60 seconds [0...240] |
| 3.4.1 | TEMP.FRIO-V PAR. | Temporize time between VSL-stop and V-stop | 10 seconds [2...60] |
| 3.4.2 | TEMP.FRIO-V ARR. | Temporize time between VSL-start and V-start | 10 seconds [2...60] |
| 3.4.3 | T.CICL.V MARCHA | Cyclic Temporize time of working fan | 0 seconds [0...240] |
| 3.4.4 | T.CICL.V PARADA | Cyclic Temporize time of stopped fan | 10 minutes [0...30] |
| 3.5.1.1 | TIPO TEMP.AMB | Ambient temperature sensor type | control nº1 → PT-100...[Do not exist]<br>rest of controls → Do not exist |
| 3.5.1.2 | AJUSTE TEMP.AMB | Ambient temperature adjustment | 0.0 [-10...+10] |
| 3.5.2.1 | TIPO HUM.AMB | Ambient humidity sensor type | Do not exist...[PT-100] |
| 3.5.2.2 | AJUSTE HUM.AMB | Ambient humidity adjustment | 0.0 [-10...+10] |
| 3.5.3.1 | TIPO TEMP.ASPIR | Suction temperature sensor type | PT-100...[Do not exist] |
| 3.5.3.2 | AJUSTE TEMP.ASP | Suction temperature adjustment | 0.0 [-10...+10] |
| 3.5.4.1 | TIPO TEMP.DESC. | Discharge temperature sensor type | Do not exist...[PT-100] |
| 3.5.4.2 | AJUSTE TEMP.DES. | Discharge temperature adjustment | 0.0 [-10...+10] |
| 3.5.5.1 | TIPO PRES.ASP. | Suction pressure sensor type | Do not exist...[4-20mA] |
| 3.5.5.2 | RANGO BAJO P.ASP. | Low rank of suction pressure | 0.0 [-1...30] |
| 3.5.5.3 | RANGO ALTO P.ASP | High rank of suction pressure | 30.0 [-1...30] |
| 3.5.5.4 | AJUSTE PRES.ASP | Suction pressure adjustment | 0.0 [-10...+10] |
| 3.5.6.1 | TIPO PRES.DESC. | Discharge pressure sensor type | Do not exist...[4-20mA] |
| 3.5.6.2 | RANGO BAJO P.DES. | Low rank of discharge pressure | 0.0 [-1...30] |
| 3.5.6.3 | RANGO ALTO P.DES | High rank discharge pressure | 30.0 [-1...30] |
| 3.5.6.4 | AJUSTE PRES.DESC | Discharge pressure adjustment | 0.0 [-10...+10] |
| 3.6.8 | RST.REPARTO FRIO | Minutes timer of cool use distribution | 0 minutes |
| 3.6.9 | CONT.HORAS FRIO | Cool hours timer | 0 hours |




# 5. CONNECTIONS OF THE CP-100 REMOTE CONTROL

The connections are made by extractable connectors, that are inserted according to the 'Connections sketch' of page num 19.

## CONNECTOR B3 (T-486-1) (POWER LINES):

Terminals 1 and 2: 220 V.A.C. 50/60 Hz.

## CONNECTOR B4 (T-486-1) (COMMUNICATION LINES):

Terminals 27 and 28: If there is a jumper between pins num. 1 and 2 of J4: RS485 Signal, to establish a serial communication with the PC.
If there is a jumper between pins num. 2 and 3 of J4: RS422 TX Signal, to establish a serial communication with the PC.

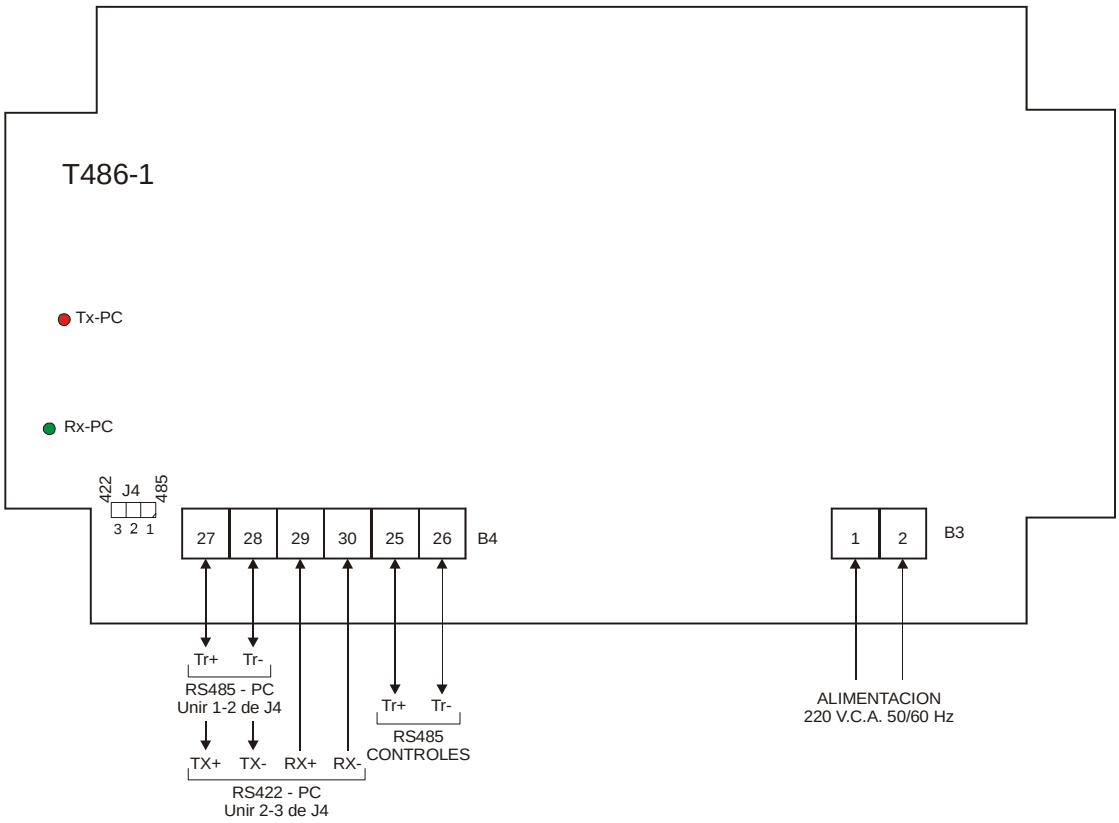
Terminals 29 and 30: If there is a jumper between pins num. 2 and 3 of J4 : RS422 RX Signal, to establish a serial communication with the PC.

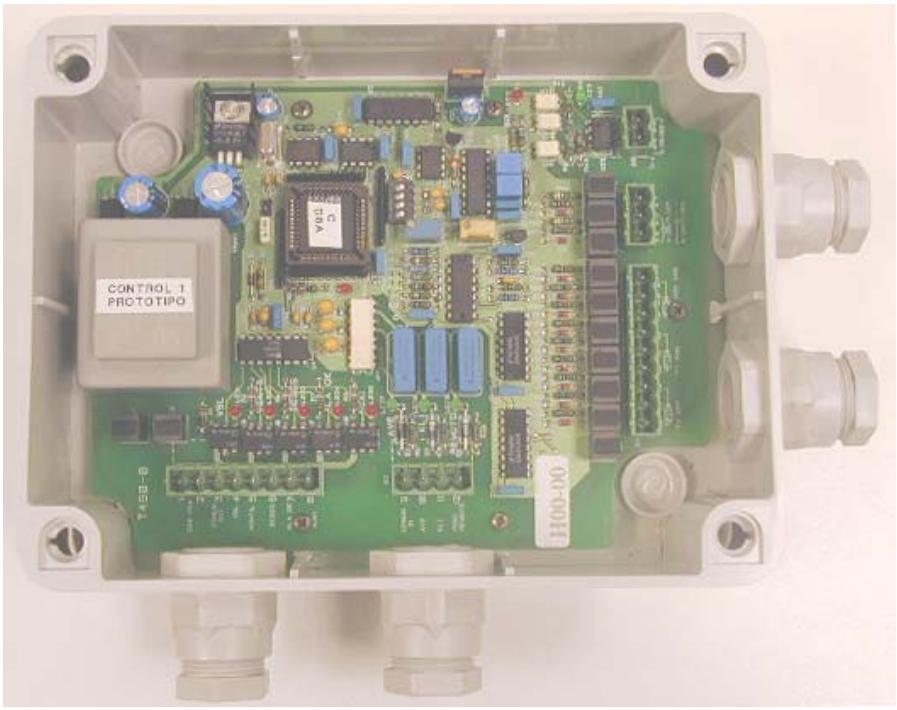
Terminals 25 and 26: RS485 Signal, to establish a serial communication with the controls.

## ADVICES FOR THE WIRE SYSTEM:

Signal and communication lines pipes should be independent from the power lines ones. There also should be the maximum separation between both pipes throughout all the power lines laying, in order to avoid interferences and electrical noises.

## COLD PACK 100 REMOTE CONTROL'S CONNECTIONS SKETCH:






# 6. COLD-PACK 100 CONTROL'S CONNECTIONS

The connections are made by extractable connectors, that are inserted according to the 'Connections sketch' of page num. 22.

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## CONNECTOR B1 (T-490-0) (POWER LINES):

Terminals 1 y 2: 220 V.A.C. 50/60 Hz.
Terminal 3: Output's common supply. Unido a una de las fases.
Terminal 4: VSL output. Relay 220 V.A.C / 5 Amp.
Terminal 5: Fan output. Relay 220 V.A.C / 5 Amp.
Terminal 6: Defrosting output. Relay 220 V.A.C. / 5 Amp.
Terminal 7: OK output. Relay 220 V.A.C / 5 Amp.
Terminal 8: Auxiliary output 1. Relay 220 V.A.C. / 5 Amp.

## CONNECTOR B2 (T-490-0) (POWER LINES):

Terminal 9: Digital input's common supply.
Terminal 10: Failure input.
Terminal 11: End-of-Defrosting Klixon input .
Borna 12: Input of remote cool order.

## CONNECTOR B3(T-490-0) (SIGNAL LINES):

Terminals 13, 14 and 15: PT100 input with three wires (Suction Temperature).

Terminals 16, 17 and 18: PT100 input with three wires (Ambient Temperature).

Terminals 19, 20 and 21: PT100 input with three wires (Ambient humidity or Discharge Temperature).

## CONNECTOR B4 (T-490-0) (SIGNAL LINES):

Terminal 22: Común de la entrada de mA (activ, > +12 V ).

Terminal 23: mA input. (Suction pressure)

Terminal 24: mA input. (Discharge pressure)

## CONNECTOR B5 (T-490-0) (COMMUNICATIONS LINES):

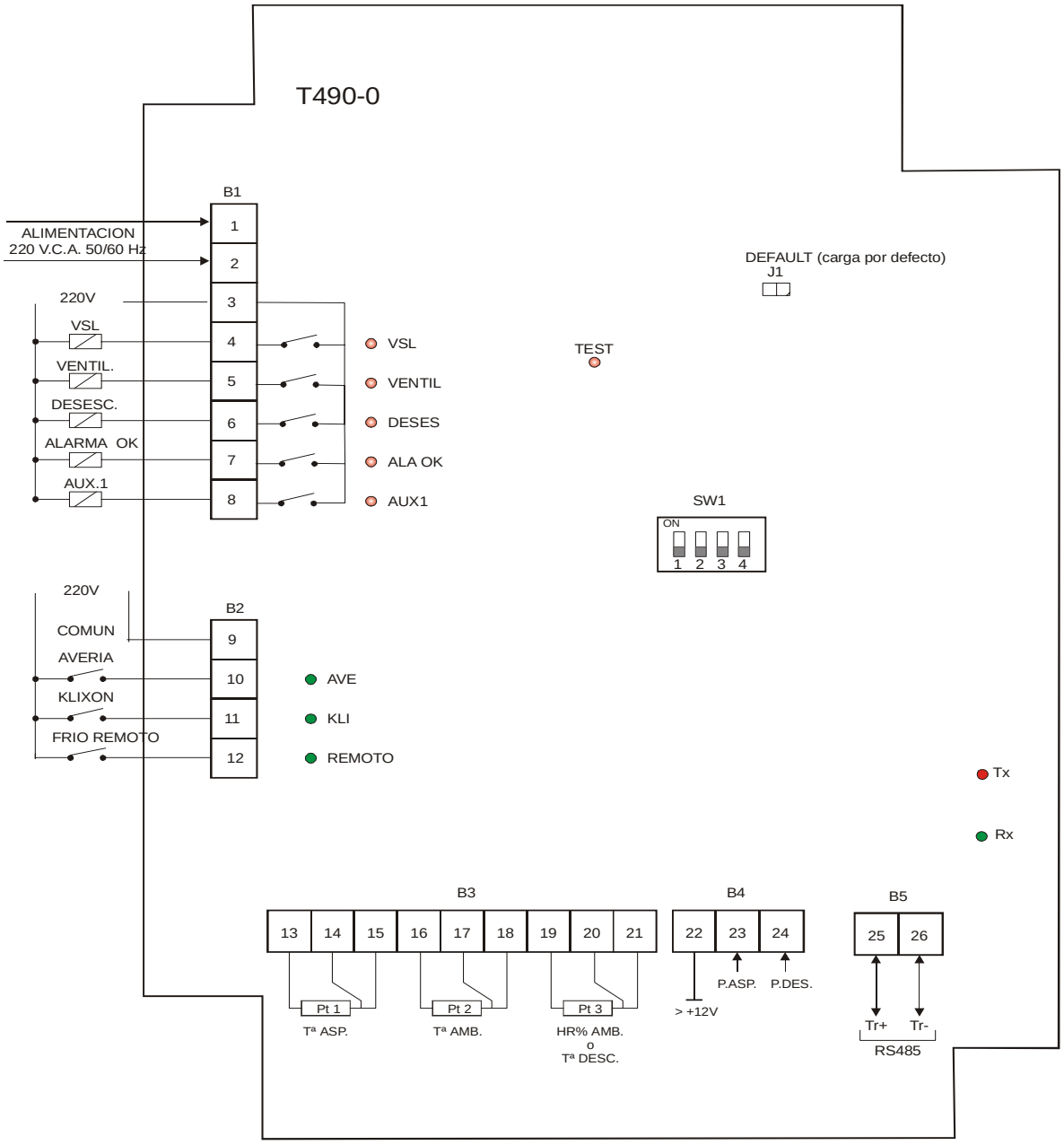
Terminals 25 and 26: RS485 Signal for serial communication.

## ADVICES FOR THE WIRE SYSTEM:

Signal and communication lines pipes should be independent from the power lines ones. There also should be the maximum separation between both pipes throughout all the power lines laying, in order to avoid interferences and electrical noises.

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# CP-100 CONTROL'S CONNECTIONS SKETCH



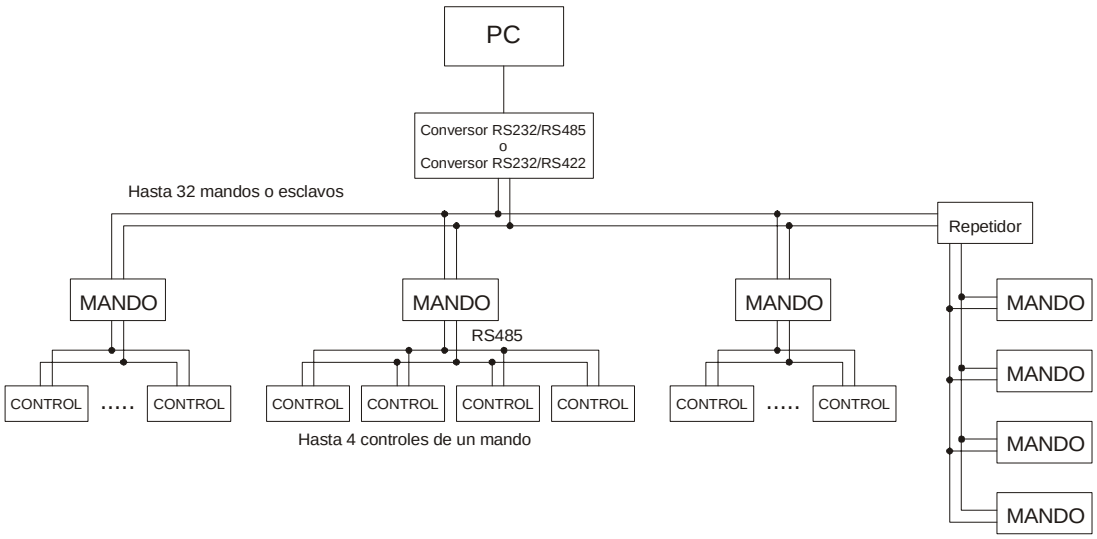
# 7. CONFIGURATION OF THE COMMUNICATION BETWEEN THE REMOTE CONTROL AND THE CONTROL

| Switch 1 - SW1 | Switch 2 - SW1 | DIRECTION / N° CONTROL |
|---|---|---|
| OFF | OFF | 1 |
| OFF | ON | 2 |
| ON | OFF | 3 |
| ON | ON | 4 |




Micro-switches installed in each control's plate.

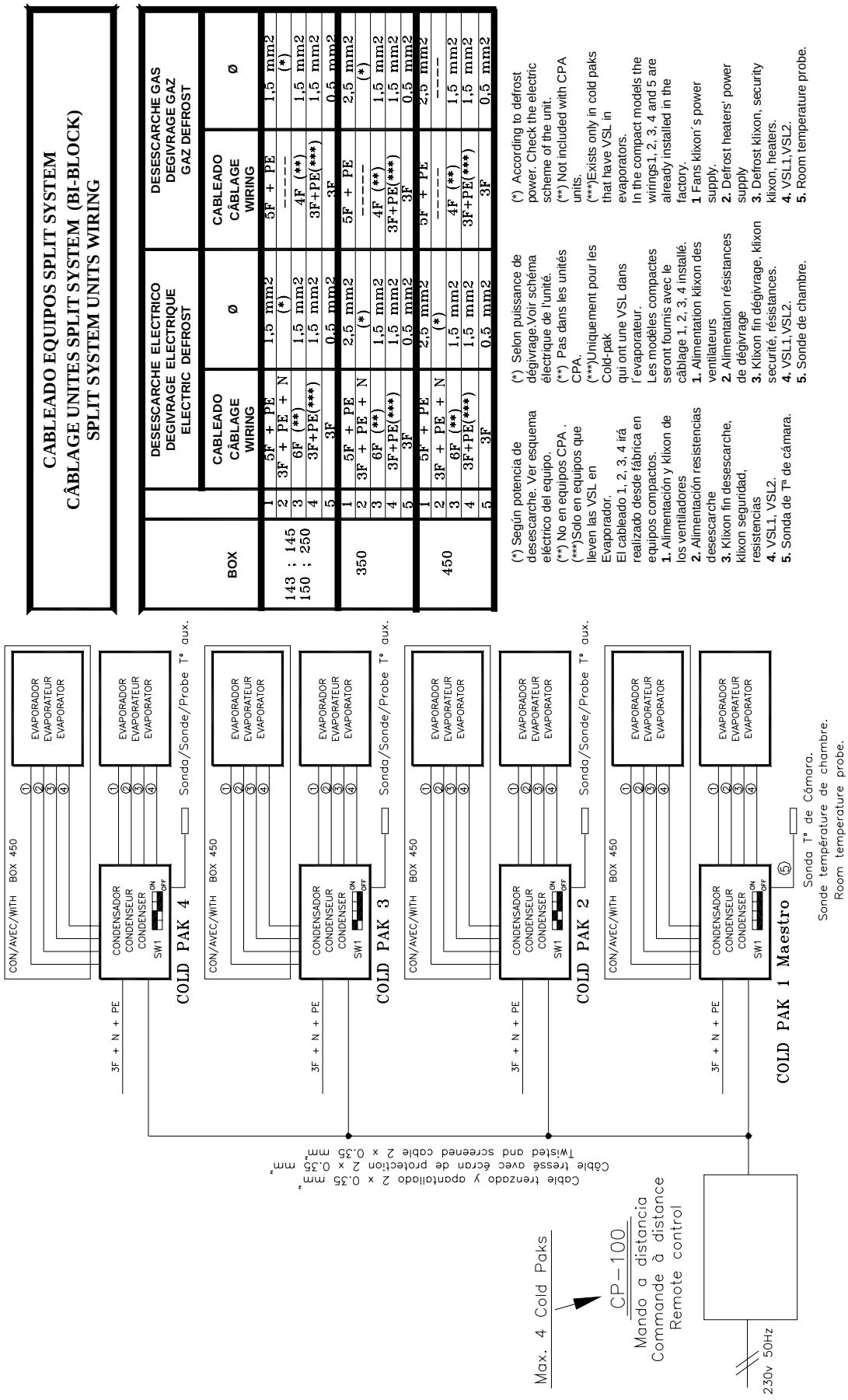
## EXAMPLE OF INSTALLATION:



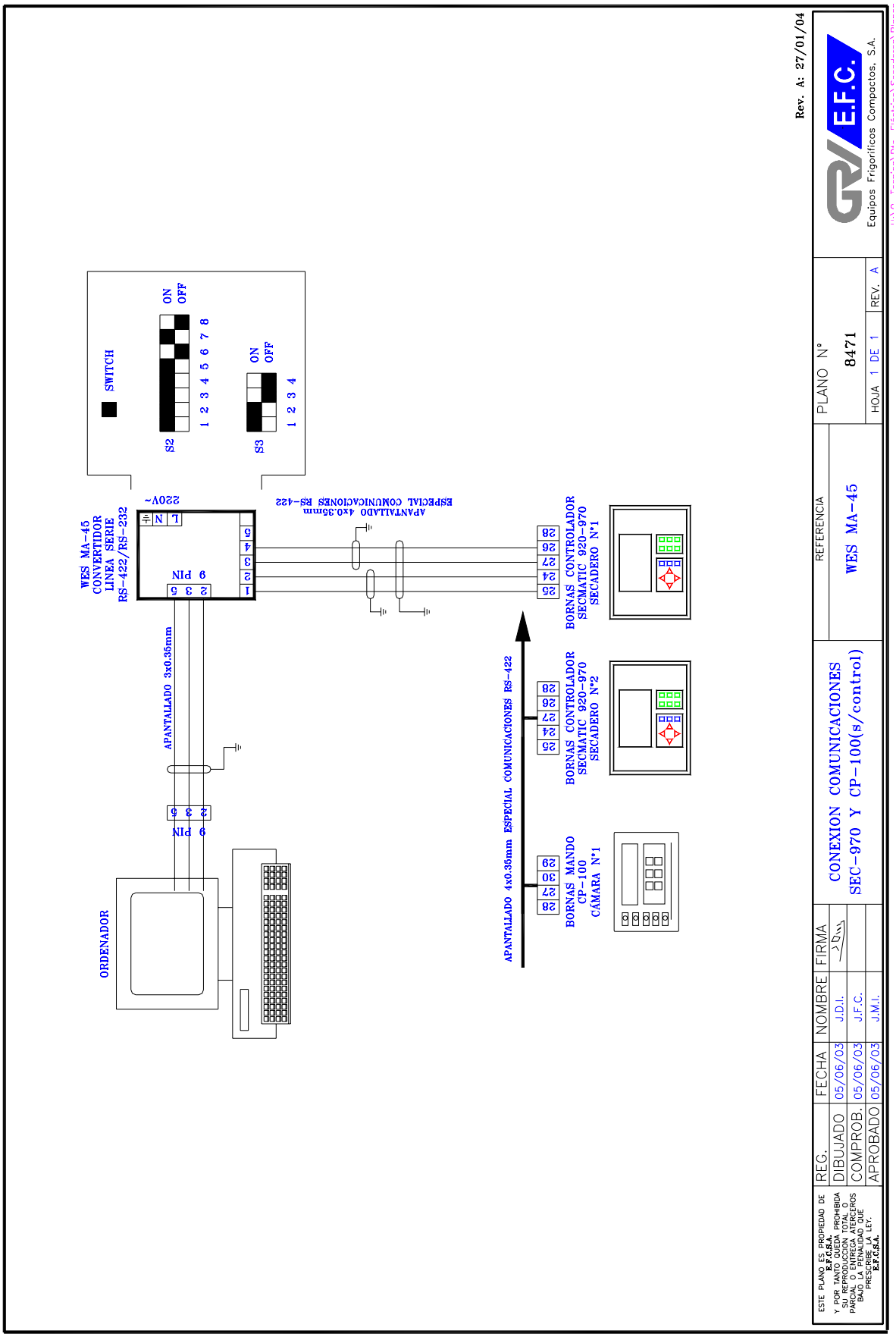
# UNIFILAR SKETCH

The evaporating unit wire of compact units is always connected at workshop.

Please, find below the sketch of a the split version.



# COMMUNICATIONS TO THE PC



# 8. ERRORS, FAILURES AND WARNINGS

Whenever there is an error, failure or warning, the display will indicate the failure type. The 'OK' output will be disconnected, depending on the seriousness of the failure. When the 'OK' output is disconnected, the failure led of the remote control will light up. The light will be intermittent, if the failure is not so important as to disconnect the 'OK' output.

As long as there is the error, failure or warning, the lower line of the display's message will be shown alternately with the normal display of each screen. A failure will be solved, as soon as its cause disappears.

The light of the Led 'TEST', which is situated in the control card, will blink each time that the control detects an error, a failure or a warning

It is possible to have several failures at the same time, but the display screen will only show the most serious one. In that case, once the biggest problem has been solved, the display screen will show a less important warning. La prioridad en los mensajes seguirá el mismo orden que la descripción de cada aviso que se realiza a continuación.

## ERRORS, FAILURES AND WARNINGS CHART

The chart below shows the errors from the most to the less important ones.

| DISPLAY'S MESSAGE DESCRIPTION | ACTIONS IN THE REMOTE CONTROL/CONTROL | POSSIBLE CAUSE | POSSIBLE SOLUTION |
|---|---|---|---|
| 'ERROR RANGE 1234'<br>The temperature set-point is out of the limits (max or min) in one or several Controls.<br>The display shows the number(s) of he control(s) that has or have caused the error. | Press any key, to delete the warning message. | - The edited temperature set-point is out of the limits, therefore his writing has been rejected. | - To check the configuration of the temperature set-point limits :<br>LIM. MIN. CONSIGNA<br>LIM. MAX. CONSIGNA |
| 'ERROR WRITE 1234'<br>A parameter or set-point has been wrong edited in one or several Controls. This error can also be found in one or several Controls, when the ON/OFF order or the Defrosting order is being executed.<br>This failure takes place, once we have tried the edition for three times in one Control.<br>The Display shows the number(s) of he control(s) that has or have caused the error. | Press any key, to delete the warning message. | - Intermittent communication failures. | - To edit again.<br>- To check the connection and the configuration of communication between the Remote Control and the Control. |
| 'ERROR COM. SLAVE'<br>Internal communication failure among the Remote Control's microprocessors: Master / Slave. | The communication with the PC wil not work. The leds do not light. If the microcontroller Slave Works, but there is not any communication between Slave and Master, all the leds will have a quick intermittent light.<br>The message will disappear as soon as the communication is re-established. | - Failure of the Remote's Control Slave microprocessor | - To switch off and then to switch on.<br>- To replace the Remote Control |

| DISPLAY'S MESSAGE<br>DESCRIPTION | ACTIONS IN THE REMOTE CONTROL/CONTROL | POSSIBLE CAUSE | POSSIBLE SOLUTION |
|---|---|---|---|
| 'ERR.COMUNICACION'<br>There is a communication failure with the control, that is displayed at that moment. | The light of the leds will be intermittent.<br>The message will disappear as soon as the communication is re-established.<br>If the communication between the controls is not correct, the main probe can not distribute the temperature and R.H. measures to the remote probes and the following elements will not work properly: the use of the compressors, time between defrostings and sequential starts.<br>If the communication error is only between a specific number of controls, the remote control will work as if there were less number of controls in the room.<br>If the control receives remote measurements, it will regulate for at least 20 seconds with the last received measure. Once the 20 seconds are over, it will stop the regulation. The Alarm output will be activated (the 'OK' out put will go down). Defrostings and cycling times will work correctly.<br>If the temperature / humidity probe is configured as "a main probe", then it will go on regulating the temperature /humidity. In this case the probe will go on working with its own temperature set-point's hysteresis value. | - There is an error in the configuration of the controls' communication.<br>- There is a failure in the hardware's communication. | - To check the connection and the configuration of the communication between the Remote Control and the Control.<br>- To check in the Control, that the switches have the correct Control address.<br>- To switch off and then to switch on the remote control and/or the control.<br>- To observe the led 'TEST' and the communications in the control card, in order to know, if they are still working. |

| DISPLAY'S MESSAGE<br>DESCRIPTION | ACTIONS IN THE REMOTE CONTROL/CONTROL | POSSIBLE CAUSE | POSSIBLE SOLUTION |
|---|---|---|---|
| 'ERR.TEMP. PRINC' or 'ERR. HUM. PRINC.'<br>There is more than one Control, that have been configured with a principal temperature or humidity probe. | The Display will show this message, each time that a Control is displayed.<br>The remote control will not distribute the measurements of the main probe.<br>The controls that have been configured with a remote probe, will also have remote temp/hum error. | - There is an error in the configuration of the Principal probe selection (local or remote). | - To check the configuration of the Controls. |
| 'ERR.TEMP. REMOTA' or 'ERR. HUM. REMOTA'<br>The temperature or humidity remote measure does not exist.<br>This error will take place, when the control is being configured with a remote probe and:<br>- It does not receive, in at least 20 seconds, the measure value.<br>- If we receive the following message = 'does not exist'. It is the case, when:<br>* the remote control detects that there is no Control that has a main probe<br>*or, if there is more that one control with main probe<br>*or, if the main probe's control has been configured as "does not exist". | The control will follow the same process as for the failures 'AVER.TEMP.REGUL.' or 'AVER. HUM.REGUL.' | -There is an error in the internal configuration. It can be that:<br>*There isn't any unit that has been configured as "main unit".<br>*Or, the main unit hasn't got any probe or has not been configured with a probe. | - To check the configuration of the controls. |

| DISPLAY'S MESSAGE<br>DESCRIPTION | ACTIONS IN THE REMOTE CONTROL / CONTROL | POSSIBLE CAUSE | POSSIBLE SOLUTION |
|---|---|---|---|
| 'AVER.ENT.EXTERNA'<br>When the failure external input has got "0 Volt". | The control will activate the alarm output (automatically, the OK output will go down) and the rest of the outp uts will also go down.<br>The External cool order is not allowed | | - To check the securities line inside the control panel. |
| 'AVER.TEMP.REGUL.'<br>There is a failure of the ambient temperature probe, that regulates the room temperature (the local or the remote probe). | The control will activate the alarm output (automatically, the OK output will go down). The humidity/dry output will also go down. The defrostings will work as usual. The cycling times will not work. | - Probe's breaking.<br>- Communication's failure. | -To check the probe and its connection or to configure another control, which has got a main temperature regulation probe .<br>- To check that the communication between the remote control and the controls is correct. To check the connection and the configuration of the CP-100. |
| 'AVER. HUM.REGUL.'<br>There is a failure of the ambient humidity probe, which regulates room humidity. (the local or the remote probe). | The humidity regulation does not work. | - Probe's breaking.<br>- Communication's failure. | - To check the probe and its connection or to configure another control, which has got a main temperature regulation probe.<br>- To check that the communication between the remote control and the controls is correct. To check the connection and the configuration of the CP-100. |
| 'AVER. TEMP. ASP.'<br>There is a failure of the suction temperature probe. | All the elements will work correctly, except if the control is configured with defrosting by hot gas. If it is the case, the defrosting will end when the "max. defrosting time" is over. | - Probe's breaking. | - To check the probe and its connection or to configure the probe as "It does not exist". |

| DISPLAY'S MESSAGE<br>DESCRIPTION | ACTIONS IN THE REMOTE CONTROL / CONTROL | POSSIBLE CAUSE | POSSIBLE SOLUTION |
|---|---|---|---|
| 'TEMPERATURA ALTA'<br>High temperature warning.<br>The temperature has gone over the set-point temperature + the hysteresis of the high temperature warning. | | | - To wait until the temperature has gone down.<br>- To check the refrigerating system's operation. |
| 'AVER. TEMP.DESC.'<br>Failure of the discharge temperature probe. | | - Probe's breaking.<br>- If the pressure probe (4/20 mA) is under 2 mA or over 22 mA.<br>- If the pressure probe has been wrong configured. That means, that the values of the maximum and minimum limits have been changed between them. | - To check the probe and its connection or to configure the probe as "It does not exist"<br>- To check the configuration. |
| 'AVER. PRES. ASP.'<br>Failure of the suction pressure probe. | | | |
| 'AVER. PRES.DESC.'<br>Failure of the discharge pressure probe. | | | |
| 'AVER.TEMP. LOCAL'<br>Failure of the local ambient temperature probe. | | | |
| 'AVER. HUM.LOCAL.'<br>Failure of the local ambient humidity probe. | | | |